

Assessment Test III

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. **Assertion (A):** At higher altitude, the height of the mercury column in a barometer decreases.

Reason (R): Atmospheric pressure decreases with an increase in altitude.

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

2. **Assertion (A):** MnO_2 is used in the preparation of O_2 from KClO_3 .

Reason (R): MnO_2 acts as a positive catalyst and decreases the melting point of KClO_3 .

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

3. Arrange the layers of atmosphere in the order of the descriptions given below.

- (i) Meteorites burn in this layer.
- (ii) This layer reflects radio signals.
- (iii) Ozone layer is present.
- (iv) Weather phenomena occur.

- | | |
|-----------------|------------------|
| (A) Ionosphere | (B) Stratosphere |
| (C) Troposphere | (D) Mesosphere |
| (a) CDAB | (b) DABC |
| (c) DCAB | (d) CDBA |

4. $\text{Na} + \text{O}_2 \rightarrow \text{product A}$

$\text{Al} + \text{O}_2 \rightarrow \text{product B}$

$\text{Fe} + \text{O}_2 \rightarrow \text{product C}$

$\text{S} + \text{O}_2 \rightarrow \text{product D}$

Study the following reactions and arrange the products in the increasing order of the number of oxygen atoms present in one molecule, considering product C as mixed oxide and product D as lower oxide.

- | | |
|----------|----------|
| (a) DACB | (b) ADBC |
| (c) ADCB | (d) DABC |

5. Match the statements of Column A with those of Column B.

Column A	Column B
(i) Rapid combustion	(A) Substance burns to produce excess amount of gas with the liberation of heat.
(ii) Spontaneous combustion	(B) Substance burns rapidly as long as the combustible substance is available.
(iii) Explosion	(C) Substance burns in air at room temperature.

(a) (i) → (C); (ii) → (A); (iii) → (B)

(b) (i) → (B); (ii) → (C); (iii) → (A)

(c) (i) → (A); (ii) → (C); (iii) → (B)

(d) (i) → (C); (ii) → (B); (iii) → (A)

6. In the process of laboratory preparation of oxygen, oxygen gas is collected by the _____.

- (a) downward displacement of water
- (b) downward displacement of air
- (c) upward displacement of water
- (d) upward displacement of air

7. Identify the true statement among the following regarding LPG and kerosene stove.

- (a) Both can be lighted by automatic gas lighter.
- (b) Both cannot be lighted by automatic gas lighter.
- (c) LPG stove can be lighted by automatic gas lighter but not kerosene stove.
- (d) Kerosene stove can be lighted by automatic gas lighter but not LPG stove.

8. Which of the following methods is used to prevent corrosion of containers that are used for storing edible materials?

- (a) Galvanization
- (b) Electroplating
- (c) Painting
- (d) Tinning

9. The major source of most of the air pollutants is _____.

- (a) industries
- (b) automobiles
- (c) thermal power
- (d) deforestation

10. Reaction between oxygen and hydrogen produces a lot of heat energy. This is exploited in which of the following uses of O_2 ?

- (a) Stimulation for breathing
- (b) As anaesthetic
- (c) Respiration
- (d) Welding

Space for rough work

Space for rough work

11. In which among the following reactions, white globules are formed?
- (a) $\text{Pb}_3\text{O}_4 \rightarrow \text{PbO} + \text{O}_2$ (b) $\text{Ag}_2\text{O} \rightarrow \text{Ag} + \text{O}_2$
(c) $\text{PbO}_2 \rightarrow \text{PbO} + \text{O}_2$ (d) $\text{HgO} \rightarrow \text{Hg} + \text{O}_2$
12. Study the following reaction and identify the metal.
- $$3\text{NH}_3 + 5\text{O}_2 \xrightarrow[\text{Metal}]{800^\circ\text{C}} 4\text{NO} + 6\text{H}_2\text{O}$$
- (a) Cr (b) Pt (c) Pd (d) Zn
13. Which among the following gases turns alkaline pyrogallol to dark brown?
- (a) O_2 (b) CO_2
(c) SO_2 (d) NO_2
14. Identify the true statement regarding the innermost zone of candle flame.
- (a) Flame is orange coloured.
(b) It is the hottest zone.
(c) It is also known as zone of unburnt wax vapours.
(d) It is the luminous zone.
15. Which among the following is not a measure to reduce global warming?
- (a) Controlling deforestation
(b) Safe disposal of industrial waste
(c) Reduction in the use of fossil fuels
(d) Restricting the use of CFCs

Assessment Test IV

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. **Assertion (A):** Mercury is used as barometric liquid.

Reason (R): Mercury can rise to a longer column.

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

2. **Assertion (A):** Preparation of O_2 from H_2O_2 has an advantage over its preparation from $KClO_3$ and $KMnO_4$.

Reason (R): H_2O_2 gives pure oxygen.

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

3. Arrange the following layers in the order of the descriptions about them given below.

- (i) Very high temperatures are present.
- (ii) Temperature of this layer is lower than other layers.
- (iii) Temperature increases up to a range of $0^\circ C - 10^\circ C$.
- (iv) Temperature decreases up to about $-50^\circ C$.

- | | |
|------------------|-----------------|
| (A) Mesosphere | (B) Ionosphere |
| (C) Stratosphere | (D) Troposphere |
| (a) BACD | (b) BADC |
| (c) ABCD | (d) ABDC |

4. The following statements are reactions of various metals with O_2 .

- (i) Reddish brown powder is formed.
- (ii) Black powder is formed.
- (iii) Slight heating is required.
- (iv) Reacts vigorously

Arrange the following metals in the order of the above observations.

- | | |
|--------------|----------|
| (A) Cu | (B) Fe |
| (C) Moist Na | (D) Ca |
| (a) BADC | (b) BACD |
| (c) ABDC | (d) ABCD |

5. Match the statements of Column A with those of Column B.

Column A	Column B
(i) Rapid combustion	(A) Burning of sodium in moist air
(ii) Explosion	(B) Burning of fuel gas
(iii) Spontaneous combustion	(C) Burning of crackers

- (a) (i) → (C); (ii) → (A); (iii) → (B)
 (b) (i) → (B); (ii) → (C); (iii) → (A)
 (c) (i) → (A); (ii) → (C); (iii) → (B)
 (d) (i) → (C); (ii) → (B); (iii) → (A)

6. Identify the true statement regarding oxygen.

- (a) O₂ is highly soluble in water.
 (b) O₂ is insoluble in water.
 (c) O₂ is slightly soluble in water.
 (d) O₂ is lighter than air.

7. Which among the following has the least ignition temperature?

- (a) Kerosene (b) LPG
 (c) CNG (d) Petrol

8. Surgical instruments and utensils do not get corroded since the material used to manufacture them is _____ for prevention of corrosion.

- (a) alloyed (b) tinned
 (c) galvanized (d) electroplated

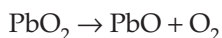
9. Which among the following pairs of pollutants are not produced by automobiles?

- (a) CFC, SO₂ (b) Fly ash, SO₂
 (c) CFC, Fly ash (d) SO₂, CO₂

10. Which of the following does not involve the usage of oxygen?

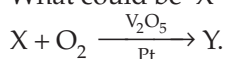
- (a) Combustion (b) Respiration
 (c) Photosynthesis (d) Manufacture of oxyacids

11. Identify the colour change in the given reaction.



- (a) Yellow to brown (b) Black to yellow
 (c) Brown to yellow (d) Yellow to black

12. What could be 'X' and 'Y' in the following reaction?



- (a) X → SO₂, Y → SO₃ (b) X → CO, Y → CO₂
 (c) X → NO, Y → NO₂ (d) X → P₂O₃, Y → P₂O₅

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13. Oxygen turns a colourless gas 'X' to reddish brown colour gas 'Y'. Identify X and Y.

- (a) $X \rightarrow \text{CO}, Y \rightarrow \text{CO}_2$ (b) $X \rightarrow \text{SO}_2, Y \rightarrow \text{SO}_3$
 (c) $X \rightarrow \text{NO}, Y \rightarrow \text{NO}_2$ (d) $X \rightarrow \text{N}_2, Y \rightarrow \text{NO}$

14. Which zone of candle flame produces soot due to incomplete combustion?

- (a) Innermost zone (b) Non-luminous zone
 (c) Outermost zone (d) Luminous zone

15. Which among the following is not an effect of global warming?

- (a) Corrosion of monuments made up of marble
 (b) Increase in frequency of floods
 (c) Unseasonal rains
 (d) Damage of vegetation

Space for rough work

Answer Keys

Assessment Test I

1. (b) 2. (c) 3. (d) 4. (a) 5. (b) 6. (a) 7. (d) 8. (a) 9. (a) 10. (d)
 11. (b) 12. (a) 13. (d) 14. (b) 15. (d)

Assessment Test II

1. (d) 2. (a) 3. (c) 4. (d) 5. (c) 6. (b) 7. (b) 8. (d) 9. (d) 10. (c)
 11. (a) 12. (a) 13. (c) 14. (a) 15. (c)

Assessment Test III

1. (a) 2. (a) 3. (b) 4. (b) 5. (b) 6. (a) 7. (c) 8. (d) 9. (b) 10. (d)
 11. (b) 12. (b) 13. (a) 14. (c) 15. (b)

Assessment Test IV

1. (c) 2. (b) 3. (a) 4. (a) 5. (b) 6. (c) 7. (c) 8. (a) 9. (c) 10. (c)
 11. (c) 12. (a) 13. (c) 14. (d) 15. (a)

Water, Solution, Solubility and Hydrogen

5



16 S Sulfur 32.066	17 Cl Chlorine 35.4527
34 Se Selenium 78.96	35 Br Bromine 79.904
51 Sb Antimony 121.757	52 Te Tellurium 127.6
	53 I Iodine 126.905

Reference: Coursebook - IIT Foundation Chemistry Class 8; Chapter - Water, Solution, Solubility and Hydrogen; Page number - 5.1–5.18

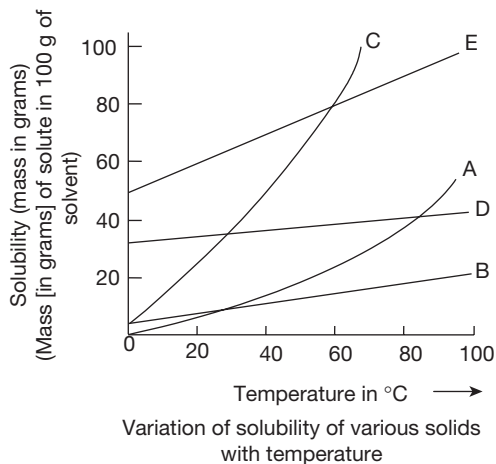
Assessment Test I

Time: 30 min.

Directions for questions from 1 to 13: Select the correct answer from the given options.

- Arrange the following changes in the increasing order of heat required. Specific heat of ice = $0.5 \text{ cal/g/}^\circ\text{C}$ and specific heat of water vapour = $0.9 \text{ cal/g/}^\circ\text{C}$
(A) 100 g ice into water at 30°C
(B) 50 g boiling water into water at 120°C
(C) 100 g boiling water into water vapour at 100°C
(D) 50 g ice into water at 0°C
(a) CABD (b) DABC (c) CBAD (d) DBAC
- Assertion (A):** Many ionic compounds show appreciable solubility in water.
Reason (R): Water has high value of dielectric constant.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.
- Identify the common feature between blue vitriol and washing soda.
(a) Both of them are efflorescent substances.
(b) Both of them have the same number of water molecules.
(c) Both of them possess water of crystallization.
(d) Both of them are coloured salts and become colourless on heating.
- Based on the following graph, arrange the substances in the increasing order of solubility at 40°C .

Space for rough work



Space for rough work

- (a) $B < A < D < C < E$ (b) $A < B < C < D < E$
 (c) $B < D < A < E < C$ (d) $A < D < B < E < C$
5. The solubility of a substance is 40 at 25°C . What amount of the substance is present in 210 g of saturated solution of that substance at the same temperature?
- (a) 82 g (b) 88 g
 (c) 60 g (d) 68 g
6. **Assertion (A):** Calcium sulphate shows same solubility at two different temperatures.
Reason (R): Water of crystallization of salt changes with change in temperature.
- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.
7. Under certain conditions, when a mixture of hydrogen and chlorine was exposed to sunlight, no reaction occurred. Which of the following reasons can be attributed to this?
- (a) Absence of moisture
 (b) Low intensity of sunlight
 (c) No reaction in presence of light energy
 (d) Presence of moisture in the reaction mixture
8. Same amount of the same sample of water is taken in four different conditions. The amount of dissolved oxygen in all cases was analysed. Identify the correct order.
- (A) Hill station in summer (B) Sea level in summer
 (C) Sea level in winter (D) Deep sea in winter

- (a) $A > B > C > D$ (b) $D > C > B > A$
 (c) $B < C < A < D$ (d) $D < C < B < A$

9. 10 g of water obtained by melting of ice is subjected to heating. What is the amount of heat required to be supplied to this water to reach its boiling point?

- (a) 4.2 J (b) 42 J
 (c) 4.2 kJ (d) 42 kJ

10. Which among the following samples when taken in equal volumes freezes at the highest temperature?

- (a) Saline water at hill station (b) Distilled water at hill station
 (c) Distilled water at plains (d) Saline water at plains

11. Match the statements of Column A with those of Column B.

Column A (Reaction)	Column B (Observation)
(i) Potassium + water	(A) Yellow substance and turns to white on cooling
(ii) Sodium + water	(B) Brown substance
(iii) Magnesium + boiling water	(C) Catches fire with lilac flame
(iv) Zinc + steam	(D) Catches fire with yellow flame
(v) Iron + steam	(E) Brilliant white light and ash

- (a) (i) → (D); (ii) → (C); (iii) → (B); (iv) → (E); (v) → (A)
 (b) (i) → (D); (ii) → (C); (iii) → (E); (iv) → (A); (v) → (B)
 (c) (i) → (C); (ii) → (D); (iii) → (E); (iv) → (A); (v) → (B)
 (d) (i) → (C); (ii) → (D); (iii) → (B); (iv) → (E); (v) → (A)

12. Although all active metals liberate hydrogen gas with acids, zinc is preferentially used in the laboratory preparation of hydrogen. Various metals are not suitable for different reasons. The metals and the reasons for their non-suitability are given below. Identify the wrong set.

- (a) Potassium - Violent explosion
 (b) Lead - Reversible reaction
 (c) Aluminium - Formation of protective coating
 (d) Magnesium - High cost

13. When a glass full of ice is subjected to melting, which of the following statements is correct?

- (a) Water overflows due to increase in density.
 (b) Water overflows due to decrease in density.
 (c) Water level in the glass decreases due to increase in density.
 (d) Water level in the glass remains constant as there is no change in density.

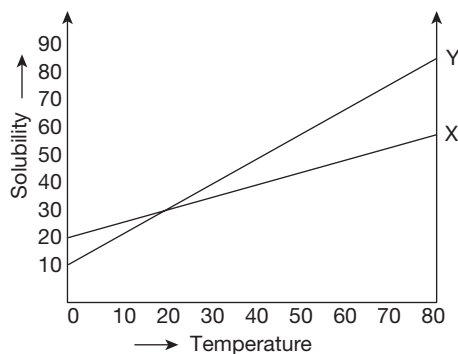
Space for rough work

5.4 Chapter 5 Water, Solution, Solubility and Hydrogen

Directions for questions 14 and 15: Answer the following questions based on the solubility curves of 'X' and 'Y' shown below and select the correct answer from the given options.

Space for rough work

14.



Two solutions of 'X' and 'Y' each containing 50 g of X and Y, dissolved in 100 mL water at 80°C temperature, are taken and cooled continuously. Below what temperatures do X and Y start crystallizing?

- (a) X → 80°C, Y → 80°C (b) X → 30°C, Y → 30°C
(c) X → 80°C, Y → 50°C (d) X → 50°C, Y → 30°C

15. What are the maximum amounts of 'X' and 'Y' that can be precipitated during the process?

- (a) 30 g, 40 g (b) 50 g, 50 g
(c) 20 g, 10 g (d) 30 g, 70 g